

Deterministic Probabilities for Thompson Sampling and Entropy Search

Peter Cotton

First version: September 1, 2026.

This version: October 6, 2026.

Abstract

Thompson sampling plays each alternative with the probability that it is the best one, and entropy search chooses its next evaluation in the same manner. Where the probabilities themselves are needed they are approximated: from marginal means and variances in bandits, and by expectation propagation in Bayesian optimization, at a cost quartic in the number of candidates. We observe that when the correlation between alternatives runs through a few shared factors, or through blocks, nested groups or a tree, these probabilities are one integral over the shared variables. Quadrature of fixed resolution then evaluates them all at once, at a cost linear in the number of alternatives. Simulation under common random numbers and a Gaussian bandit with a factor prior produce such beliefs exactly, and a dense covariance is fitted to the form on intake. With the computed probabilities a stopping rule stops sooner at the same nominal threshold, sampling from them reproduces Thompson sampling’s regret, and entropy search runs on thousands of candidates rather than fifty.

1 Introduction

An analyst has twenty candidate inventory policies and estimates each one’s expected profit by simulation. A website has fifty advertisements and records clicks on whichever one it shows. A chemist has predicted binding strengths for a thousand candidate molecules. In each case there are n alternatives, and alternative i has an unknown value θ_i : the policy’s true expected profit, the advertisement’s true click rate, the molecule’s true binding strength. The data $\mathcal{D}$ collected so far give a posterior belief about $\theta = (\theta_1, \dots, \theta_n)$.

The posterior probability of optimality is the vector

$$p_i = \Pr\left(\theta_i = \max_j \theta_j \mid \mathcal{D}\right), \quad i = 1, \dots, n. \quad (1)$$

Its entry p_i is the probability, given the data, that alternative i is truly the best. If $p_3 = 0.9$ for the inventory policies, there is a ninety percent chance that policy 3 has the highest expected profit.

Decisions about where to collect data use this vector. The analyst can stop simulating once the largest p_i exceeds 0.95, and until then can spend runs on the policies whose p_i are close. The website can show advertisement i with probability p_i , which is Thompson sampling. A rule that must respect constraints, or must be reproducible without randomness, needs the numbers p_i themselves.

Thompson sampling runs in production wherever alternatives are compared under uncertainty: advertising, recommendation, clinical allocation, and now the fine-tuning of language models toward it (Menet et al., 2026). The policy never needs the numbers p_i , but much that surrounds the policy does.

Off-policy evaluation needs them as propensities, and an audit of why an arm was shown needs them. A stopping rule, a shortlist of the k arms most likely to be best, a constraint on how often any arm may be shown, and a reproducible version of the policy all need them. In every one of these uses they have been approximated, from marginals or by counting posterior draws.

When the beliefs about different alternatives are independent, each p_i is a one-dimensional integral. In practice they are usually correlated, because alternatives share causes. Inventory policies simulated on the same demand history all do well on a quiet day and badly on a busy one. Advertisements aimed at the same audience have similar click rates. Molecules built on the same scaffold bind similarly.

With correlated beliefs, p_i is the probability that the $n - 1$ differences $\theta_i - \theta_j$ are all positive, a Gaussian orthant probability in $n - 1$ dimensions. No general fast method computes it, and until recently none computed it quickly even when the correlation comes from a few shared factors, which is the case the pricing pass of Cotton (2026) now covers.

Three literatures need this vector and approximate it. The first is ranking and selection, which chooses the best of n simulated systems. Its procedures bound the probability of a correct selection with Bonferroni-type inequalities, or estimate it by posterior sampling. The joint vector is not evaluated.

The second is Bayesian bandits and reinforcement learning. Tarbouriech et al. (2023) call Equation (1) the key object of that field and write that computing it “involves computing several complicated integrals with respect to the posterior and is intractable in most cases.” Their method, VAPOR, replaces it with a variational program. Menet et al. (2026) fine-tune a large language model toward a policy that uses only the marginal means and standard deviations of the posterior.

The third is Bayesian optimization. Entropy search (Hennig and Schuler, 2012) takes the vector itself, the posterior probability that the optimum lies at each of n candidate points, as what an optimizer should learn, and chooses each evaluation to reduce its entropy. Hennig and Schuler (2012) call the vector intractable, approximate it by expectation propagation at a cost of $O(n^4)$, and run with fifty candidates. Expectation propagation accepts any covariance.

Correlation that comes from a few shared causes has a simple form. Each value is its own part plus a weighted sum of ρ shared variables:

$$\theta_i = m_i + w_i'z + \varepsilon_i, \quad z \sim N(0, I_\rho), \quad \varepsilon_i \sim N(0, d_i), \quad (2)$$

with all the ε_i independent of each other and of z . For the inventory policies, z might be the overall demand level and its volatility. The row w_i says how strongly policy i responds to each. The covariance of θ is then

$$\Sigma = WW' + \text{diag}(d), \quad W \in \mathbb{R}^{n \times \rho}, \quad \rho \ll n, \quad (3)$$

where W has rows w_i' . The variables in z are called factors, and W holds the loadings.

The best-known Thompson samplers already have this form. Linear Thompson sampling (Agrawal and Goyal, 2013), the version Chapelle and Li (2011) evaluated on display advertising and news recommendation, puts a Gaussian posterior on a coefficient vector β of dimension equal to the number of features, and the arm means $\theta_i = x_i'\beta$ then have covariance $X\Sigma_\beta X'$. That is

Equation (3) with the feature matrix as the loadings and a factor rank equal to the number of features, and Section 3.2 keeps it in that form after every pull.

For beliefs of this form, the whole vector is one integral over the factors, and quadrature evaluates it for all n alternatives at once in time linear in n at fixed resolution. The computation is the pricing pass for correlated contests in Cotton (2026), restated rather than derived. One standard model from each of the first two literatures produces a posterior of exactly this form after every update: simulation under common random numbers, and a Gaussian bandit with a factor prior. In Bayesian optimization the form is a modelling choice, a kernel with finitely many features, or the intake fit of a dense kernel.

In the simulation, a stopping rule on the computed vector stops sooner than rules on the marginal approximations at the same nominal threshold. In the bandit, sampling from it incurs less regret than they do. In Bayesian optimization, entropy search can run on thousands of candidates rather than fifty.

2 The computation, restated from Cotton (2026)

The computation is the correlated-contest pricing pass of Cotton (2026), restated in the notation of Equation (2) and specialised to Gaussian noise. Nothing in it is new.

The original is more general: it accepts any base distribution, and a covariance given as factor loadings, as blocks, as nested groups or as a tree, or a dense covariance fitted to one of those structures on the way in, with the residual of that fit reported as its error. It also inverts the vector, recovering means from probabilities. For every structure in that grammar the cost is linear in n at fixed resolution. The factor form of Equation (3) with Gaussian noise is the case needed here.

Write the posterior in the form of Equation (2), $\theta = m + Wz + \varepsilon$. Assume every $d_i > 0$, so the maximum is almost surely unique and the p_i sum to one.

The shared factors z make the alternatives dependent. Given z , nothing is shared, and the alternatives are independent with means $m_i + w'_i z$ and variances d_i . The probability that i is best given z is then a one-dimensional integral, and p_i is its average over z .

Let $F_{i|z}$ and $f_{i|z}$ be the conditional distribution and density functions of θ_i given z . Alternative i is the maximum when every other alternative falls below it, so

$$p_i = \int_{\mathbb{R}^\rho} \phi_\rho(z) \int_{\mathbb{R}} f_{i|z}(x) \prod_{j \neq i} F_{j|z}(x) dx dz. \quad (4)$$

The inner integrands for different i share the product $G_z(x) = \prod_j F_{j|z}(x)$. Dividing out the i th term gives every p_i from the same product:

$$p_i = \mathbb{E}_z \int f_{i|z}(x) \frac{G_z(x)}{F_{i|z}(x)} dx. \quad (5)$$

On a lattice of L points in x , with Q quadrature nodes in z , the whole vector costs $O(nLQ)$. The accuracy depends on L , on the lattice window and on Q , and the Q needed grows with the number of factors and the geometry of W .

Only differences between loadings matter. Let $P = I - \mathbf{1}\mathbf{1}'/n$ and write $Wz = (PW)z + \mathbf{1}(\bar{w}'z)$, where $\bar{w}$ is the average row of W . The second term shifts every alternative by the same amount and changes no ranking, so $p(m, W, d) = p(m, PW, d)$. The number of factor dimensions the

integral has to resolve is the rank of PW , which centering can reduce below the rank of W . Centering, and any orthogonal rotation of the columns of W , leave the computed vector unchanged to 10^{-11} or better, which is a check on an implementation. In the aligned regime of the stopping experiment every row of W is the same, so $PW = 0$ and the contest is independent.

The integral is exact. Its evaluation is not, because the lattice in x and the quadrature rule over z have finite resolution. The lattice is converged, and the factor rule is accurate unless correlations are extreme. For three alternatives with $m = 0$, $W = (0, 1, 3)'$ and $d_i = 0.1$, two of the three correlations exceed 0.9, the exact vector is known in closed form from the bivariate quadrant identity, and the default rule is off by 0.0015 in total variation. With 2^{15} scrambled-Sobol nodes in place of the default rule that error is gone. The experiments' correlations are at most 0.75, and on their posteriors the default rule's error is about 10^{-6} per entry.¹

3 Two models whose posteriors have exact factor form

The computation of Section 2 needs the posterior covariance in the form of Equation (3). Two standard models produce it exactly after every update, one from ranking and selection and one from bandits. In simulation, the shared factors come from running every system on the same random inputs. In the bandit, they come from a prior in which arms share features.

3.1 Simulation with common random numbers

The result of this subsection is simple to state. A simulation study runs each of n systems many times, and each run is a replicate. When every replicate feeds the same random inputs to all n systems, the uncertainty about the n unknown means is correlated, and the derivation shows that this correlation has the same shape as the shared inputs: a few common factors plus independent noise, with as many factors as there are shared input streams. Accumulating replicates sharpens the posterior but never changes that shape. So the vector of Section 2 can be computed after every replicate in time linear in n , and the $n \times n$ posterior covariance is never formed.

A simulation study compares n systems, such as the inventory policies of the introduction, by running each one many times and averaging its output. The unknown θ_i is system i 's long-run average output.

A standard way to compare systems more precisely is to give every system the same random inputs in each replicate. This is called common random numbers. In replicate r , all twenty inventory policies face the same simulated demand history. A high-demand history raises every policy's profit, so the outputs move together, and the differences between policies are measured more precisely than their levels.

Summarise the shared random inputs of replicate r by ρ variables F_r , such as the demand level and its volatility. The analyst does not record F_r . System i responds to it with loadings v_i , which are known, and has its own noise ε_{ir} :

$$Y_{ir} = \theta_i + v_i' F_r + \varepsilon_{ir}, \quad F_r \sim N(0, I_\rho), \quad \varepsilon_{ir} \sim N(0, \sigma_i^2). \quad (6)$$

¹Measured on ten posteriors, two from each regime of each experiment. Raising the lattice from 257 to 2,049 points changes no entry by more than 4×10^{-16} . Replacing the default factor rule by 2^{15} scrambled-Sobol nodes changes no entry by more than 1.2×10^{-6} . Four million Monte Carlo draws differ from the computed vectors by 1 to 4×10^{-4} per entry, their own standard error, and the 2×10^5 draws taken at sixteen posteriors during the experiments differ by at most 0.0042 in total variation, again their own noise. These are measurements, not error bounds. The script is `research/rs_crn/exp3.quadrature`.

The F_r and ε_{ir} are independent of each other and across replicates. The loadings v_i and noise scales $\sigma_i > 0$ are known properties of the simulation, and F_r is latent. If F_r were recorded too, subtracting $v_i' F_r$ from each output would remove the correlation and leave a diagonal posterior. The unknown is θ , with prior $\theta \sim N(0, s_0^2 I)$ and $s_0 > 0$.

The posterior covariance of θ has factor form, with the same number of factors ρ . The derivation takes four steps.

The covariance of one replicate across systems is

$$\Lambda = VV' + D, \quad D = \text{diag}(\sigma_1^2, \dots, \sigma_n^2). \quad (7)$$

After $n_r \geq 1$ replicates with mean vector $\bar{Y}$, the posterior of θ is Gaussian with precision

$$A = s_0^{-2} I + n_r \Lambda^{-1}. \quad (8)$$

The first step inverts Λ with the Woodbury identity:

$$\Lambda^{-1} = D^{-1} - D^{-1} V M^{-1} V' D^{-1}, \quad M = I_\rho + V' D^{-1} V. \quad (9)$$

With $U = D^{-1} V$ and $S = n_r M^{-1}$, the precision becomes

$$A = \text{diag}(a) - U S U', \quad a_i = s_0^{-2} + n_r \sigma_i^{-2}. \quad (10)$$

This is a positive diagonal minus a positive semidefinite term of rank at most ρ .

The second step inverts A , again with the Woodbury identity:

$$A^{-1} = \text{diag}(a)^{-1} + \text{diag}(a)^{-1} U C U' \text{diag}(a)^{-1}, \quad C = (S^{-1} - U' \text{diag}(a)^{-1} U)^{-1}. \quad (11)$$

The third step shows that C is positive definite, so that it has a square root. Let $K = \begin{pmatrix} \text{diag}(a) & U \\ U' & S^{-1} \end{pmatrix}$. The Schur complement of S^{-1} in K is $\text{diag}(a) - U S U' = A$, which is positive definite, and S^{-1} is positive definite, so K is positive definite. The Schur complement of $\text{diag}(a)$ in K is C^{-1} , so C^{-1} is positive definite too.

The fourth step gives the factor form:

$$\Sigma_{\text{post}} = \text{diag}(d) + W W', \quad d_i = a_i^{-1}, \quad W = \text{diag}(a)^{-1} U C^{1/2}, \quad (12)$$

exactly, for every $n_r \geq 1$. Before any replicate the posterior is the diagonal prior. The posterior mean is $m = A^{-1} h$ with $h = n_r \Lambda^{-1} \bar{Y}$, and both factors are available from Equations (9) and (11).

No step approximates. The posterior has factor form because the shared random inputs have factor form. The number of posterior factors is at most ρ , the number of shared input variables, however many systems are compared.

3.2 A bandit whose arms share features

The result here is the same as for the simulation. When the prior on the arms' mean rewards has a few common factors plus independent noise, the posterior keeps that shape after every pull, with no more factors than the prior, whatever arms are pulled and however often.

In a bandit, a decision maker repeatedly picks one of n arms and observes a noisy reward from it. Here θ_i is arm i 's mean reward, such as an advertisement's click rate. Pulling arm a returns $\theta_a + N(0, \sigma_n^2)$, with $\sigma_n > 0$.

The correlation comes from the prior. Advertisements that share an audience or a design are expected to have similar click rates. A linear model expresses this: arm i has known features v_i , and $\theta_i = v_i' \beta + e_i$, with a shared $\beta \sim N(0, I_\rho)$ and independent $e_i \sim N(0, d_{0i})$, $d_{0i} > 0$. The prior is then $\theta \sim N(0, \Sigma_0)$ with $\Sigma_0 = VV' + \text{diag}(d_0)$, which has factor form.

A Gaussian-process prior with a kernel of finitely many features and an independent residual has this form exactly. A dense kernel does not, and an inducing-point approximation to one has a dense residual (Titsias, 2009), so for those a diagonal residual is an added approximation.

With per-arm pull counts c , the posterior precision is

$$A = \Sigma_0^{-1} + \sigma_n^{-2} \text{diag}(c). \quad (13)$$

Equation (9) applied to Σ_0 puts A in the form of Equation (10), with $a_i = d_{0i}^{-1} + c_i \sigma_n^{-2}$, $S = M_0^{-1}$ and $M_0 = I_\rho + V' \text{diag}(d_0)^{-1} V$. The remaining steps of Section 3.1 apply unchanged. The posterior is exactly factor plus diagonal after every pull, with no more factors than the prior.

The same posterior has a form with no subtraction inside an inverse. Let $\lambda_i = c_i / \sigma_n^2$ be the observation precision accumulated on arm i , $r_i = 1 / (1 + d_{0i} \lambda_i)$, $R = \text{diag}(r)$ and $H = I_\rho + V' \text{diag}(\lambda_i r_i) V$. Then

$$\Sigma_{\text{post}} = \text{diag}(d_{0i} r_i) + R V H^{-1} V' R, \quad W_{\text{post}} = R V H^{-1/2}. \quad (14)$$

H is the identity plus a positive semidefinite matrix, so it is positive definite. An arm never pulled has $\lambda_i = 0$, and arms observed with different noise levels enter through their own λ_i . Equation (14) agrees with direct inversion of the dense precision to machine precision, and is the form to implement.

4 Thompson sampling and the approximations of the vector

Bandit methods relate to the vector in two ways. Thompson sampling uses it implicitly, without computing it. Two other methods compute approximations of it from marginal means and variances. The experiments compare all three with the computed vector.

4.1 Thompson sampling selects arm i with probability p_i

Thompson sampling draws one sample $\tilde{\theta}$ from the posterior and pulls the arm with the largest sampled value. The probability that it pulls arm i is the probability that arm i is largest under the posterior, which is p_i . Lemma 8 of Tarbouriech et al. (2023) states the general version: the expected occupancy of Thompson sampling equals the posterior probability of optimality.

Pulling an arm with probability p_i , called *exact matching*, is therefore Thompson sampling in distribution. Pulling from the computed vector $\hat{p}$ is Thompson sampling up to the quadrature error, which is about 10^{-6} per entry on the bandit's posteriors. The bandit experiment compares the two. The policy remains randomized. The deterministic part is the evaluation of its probabilities.

The computed vector gives the numbers themselves, which a single draw does not. A stopping rule needs $\max_i p_i$, an allocator needs to know which p_i are close, and a reproducible policy needs the probabilities it would otherwise sample.

4.2 Approximations from marginal means and variances

When the numbers p_i are needed, current methods approximate them using only each alternative’s marginal mean m_i and marginal standard deviation $s_i = \sqrt{d_i + \|w_i\|^2}$. Both ignore the correlation between alternatives.

F-LITE (Menet et al., 2025) treats the alternatives as independent, $\prod_i N(m_i, s_i^2)$, and approximates the probability of being best by $\Phi((m_i - \kappa^*)/s_i)$, with a threshold κ^* chosen so that these sum to one. The threshold step is a second approximation. With two independent arms, $N(1, 1)$ and $N(0, 1)$, it gives 0.691 for the first arm, against the exact 0.760. F-LITE therefore departs from p even when the posterior is independent.

Discarding correlation can produce a much larger error, because the marginals do not determine the vector, even within the covariance class of Equation (3). Let $\theta_1 = E_1$ and $\theta_i = \sqrt{1 - \epsilon} Z + \sqrt{\epsilon} E_i$ for $i = 2, \dots, n$, with Z and the E_i independent standard Gaussians. Every alternative has the same $N(0, 1)$ marginal, the covariance is rank one plus a positive diagonal, and every correlation is nonnegative. As $\epsilon \rightarrow 0$ the last $n - 1$ alternatives become copies of one another, so $p_1 \rightarrow 1/2$ and every other $p_i \rightarrow 1/(2(n - 1))$. A method that uses only the marginals returns $1/n$ for every alternative, since the marginals are identical. With $n = 100$ and $\epsilon = 10^{-4}$, $p_1 = 0.493$ against 0.01, and the total-variation error of the marginal answer tends to $1/2 - 1/n$.

A second example fixes the mean gap and varies the size of a shared shock. Let $\theta_1 = 3 + Z + \varepsilon_1$ and $\theta_2 = Z + \varepsilon_2$, with $Z \sim N(0, \tau^2)$ and independent $\varepsilon_i \sim N(0, 1)$. The shared Z cancels from the difference, so $p_1 = \Phi(3/\sqrt{2}) \approx 0.983$ for every τ . F-LITE sees only the marginal variances $1 + \tau^2$ and gives $\Phi(3/(2\sqrt{1 + \tau^2}))$, which is 0.933 at $\tau = 0$, 0.682 at $\tau = 3$, and tends to one half as τ grows. The aligned regime of the stopping experiment is this example with a hundred systems.

The shared product in Equation (4) has a precedent in the independent case. Menet et al. (2025) write the independent probability of being best as the same product integral, their Equation (2), and their Proposition 1 evaluates it by quadrature on shared integration points with a convergence guarantee. Equation (5) is that evaluator conditional on the factors, averaged over them. Their A-LITE and F-LITE are approximations of the independent integral: A-LITE fits a Gaussian to the distribution of the largest competitor by matching quartiles, and F-LITE replaces it by the shared threshold above. The experiments include F-LITE only.

The *VBOS policy* is the variational Bayesian optimistic sampling rule of O’Donoghue and Lattimore (2021), extended by Tarbouriech et al. (2023). In the near-closed form of Menet et al. (2026) it is

$$\tilde{\pi}_i = v\left(\frac{m_i - \kappa^*}{s_i}\right), \quad v(c) = \exp\left(-\frac{1}{8}(\sqrt{c^2 + 4} - c)^2\right), \quad (15)$$

with κ^* again chosen so the vector sums to one.

The VBOS policy departs from p on purpose, and its regret analysis depends on that departure. It is a policy, not an estimate of p . Its distance from p matters because Menet et al. (2026) use it as the trainable substitute for p .

5 First experiment: when a simulation study can stop

The first experiment runs the simulation study of Section 3.1, in which n systems are simulated on shared random inputs, and asks when it can stop. A simulation study stops when it is confident

enough that it has found the best system. The experiment asks whether a stopping rule that uses the computed vector stops earlier than rules that use the marginal approximations, and whether it still selects the best system as often as its confidence level promises. Correlation is the only thing that separates the rules, so everything else is held fixed and only the correlation geometry varies.

The setting is the posterior of Section 3.1 with $n = 100$ systems, $\rho = 2$ scenario streams and prior scale $s_0 = 1$. True means are drawn from the prior. Every system has replicate variance one, so each replicate carries the same marginal information in every regime, and a marginal-only rule faces the same problem in every regime. The *factor share* is the fraction of that variance that comes from the shared scenario.

Correlation can mislead a marginal-only rule in two directions. If the leading systems move together, their difference is less uncertain than their marginals suggest, and the rule is too cautious. If they move oppositely, their difference is more uncertain, and the rule is too confident. The regimes produce each case, with an uncorrelated control.

- *Independent*: factor share zero. This is the control.
- *Aligned*: every system loads on the same stream, with factor share 0.6. Differences between systems then vary less than their marginal variances suggest.
- *Opposed*: half the systems load positively and half negatively on one stream, with factor share 0.6. Differences between systems in opposite halves then vary more than their marginal variances suggest.

The experiment compares three stopping rules, which differ only in how they estimate p . The exact rule uses the computation of Section 2, and the other two use F-LITE and the VBOS policy. After every replicate, a rule recomputes its estimate of p , and it stops the first time the largest entry reaches $1 - \delta$, with $\delta = 0.05$, selecting that entry.

A fourth rule is the one a practitioner would reach for: draw M samples from the correlated posterior and estimate p by the argmax frequencies. It uses the correlation, costs $O(Mn\rho)$, and needs M of about 500 to resolve a threshold of 0.95 to within 0.01, with the error growing as the threshold is approached. It is not among the rules run, which are the two marginal rules from the literature.

For the true vector the rule has a Bayesian guarantee. Let τ be the first time $\max_i p_i \geq 1 - \delta$, let B be the budget, and on $\{\tau \leq B\}$ let I be the selected alternative and I^* the best. Then

$$\Pr(I \neq I^* \mid \tau \leq B) = \mathbb{E}[1 - p_I(\mathcal{D}_\tau) \mid \tau \leq B] \leq \delta. \quad (16)$$

Optional stopping does not affect this statement, which averages over the prior. It is not a frequentist guarantee for a fixed θ , and it applies to the true p . For the computed vector $\hat{p}$, if $\|\hat{p} - p\|_\infty \leq \eta$ at stopping, then stopping at $\max_i \hat{p}_i \geq 1 - \delta$ gives error at most $\delta + \eta$, and a threshold of $1 - \delta + \eta$ restores δ . On the stopping posteriors, η is about 10^{-6} by the measurements of Section 2, which are measurements rather than a bound.

The three rules share a nominal threshold, not a true confidence level. With two independent arms of equal posterior standard deviation s , an entry of 0.95 requires a mean gap of 2.33 s for the exact vector, 3.29 s for F-LITE and 4.84 s for the VBOS policy. The true posterior probability of a correct selection at those gaps is 0.950, 0.990 and 0.9997. Part of any extra sampling by the marginal rules is this higher effective target.

regime	rule	mean replicates	realized PCS	censored of 200
independent	exact	141.2	0.970	35
	F-LITE	207.0	0.985	65
	VBOS	271.5	1.000	102
aligned	exact	69.7	0.958	9
	F-LITE	175.6	1.000	60
	VBOS	240.1	1.000	91
opposed	exact	106.3	0.944	21
	F-LITE	171.7	0.979	57
	VBOS	239.0	0.991	88

Table 1: Stopping at nominal 95% confidence. Each regime has 200 replications and a budget of 400 replicates. Mean replicates is the mean of the stopping time capped at 400, so a censored run counts 400. Realized PCS is the fraction of stopped runs that selected the true best system. The stopped runs differ by rule.

In Table 1, every rule’s realized PCS among its stopped runs is at or above nominal, within binomial noise. The exact rule’s 0.944 in the opposed regime has 179 stopped runs and a standard error of 0.016.

At this common nominal threshold and budget, the two marginal-only rules used 1.5 to 3.4 times the capped mean replicates of the exact rule, and hit the budget far more often. The gap is largest in the aligned regime, where the leading systems’ differences are tightly coupled. There the exact rule stops after 69.7 replicates on average, against 175.6 and 240.1, and stops in 191 of 200 runs, against 140 and 109. The thresholds are nominal and not calibrated to a common PCS, so these ratios compare spend at a common threshold, not at a matched true error.

The marginal-only estimates are far from the exact vector along the way. Before stopping in the aligned regime, F-LITE is at median total-variation distance 0.123 from the exact vector, with ninetieth percentile 0.174. The VBOS policy is at 0.141, with ninetieth percentile 0.206.

The distances do not vanish in the independent regime, where they are 0.063 and 0.096. F-LITE’s threshold step is biased even when the posterior is independent, and the VBOS policy departs from p by design. At the stopping experiment’s Monte Carlo checks, the largest total-variation distance of the computed vector was 0.0015.

6 Second experiment: a bandit with correlated arms

The second experiment runs the bandit of Section 3.2, in which arms share features through a factor prior, and compares the four policies of Section 4. It asks two things. The first is whether pulling arms with the computed probabilities reproduces the regret of Thompson sampling, as Lemma 8 requires of the true vector. The second is how much regret the two marginal approximations lose when their probabilities are used as the policy.

Thompson sampling here draws from the full correlated posterior, so it already uses every correlation the computed vector uses. It is the benchmark the computed vector has to match, not a baseline it is expected to beat. The policies that lose are the two marginal approximations.

The setting is the posterior of Section 3.2 with $n = 50$ arms, prior variance one per arm,

observation noise $\sigma_n = 1$, horizon 200 and 100 replications. The prior’s factor term has two columns. In the *independent* regime it is zero. In the *clusters* regime each half of the arms loads on its own factor, which carries 0.75 of the prior variance.

The experiment compares four policies on the same posterior: Thompson sampling, exact matching, sampling from the VBOS policy, and sampling from F-LITE. Table 2 reports cumulative Bayes regret and, at the horizon, whether the arm with the highest posterior mean is the true best and the shortfall of that arm’s value below the best. Its last column is the total-variation distance between each policy’s action distribution and the exact vector at mid-horizon, along that policy’s own trajectory.

regime	policy	regret	identification	shortfall	TV to exact p
independent	Thompson	172.1	0.72	0.073	0
	exact matching	173.9	0.75	0.055	0
	VBOS	190.3	0.76	0.061	0.159
	F-LITE	179.2	0.79	0.052	0.057
clusters	Thompson	122.7	0.61	0.097	0
	exact matching	124.8	0.51	0.113	0
	VBOS	135.9	0.59	0.105	0.129
	F-LITE	128.7	0.51	0.121	0.061

Table 2: Mean over 100 replications of cumulative Bayes regret, the rate at which the highest posterior mean identifies the true best arm at the horizon, and the shortfall of that arm. The last column is the total-variation distance from the computed vector at mid-horizon. It is zero for exact matching by construction. For Thompson sampling it is zero relative to the true p by Lemma 8, and is entered as zero rather than measured.

Sampling from the true p has the same expected regret as Thompson sampling, by Lemma 8, and the computed vector is within about 10^{-6} of the true p on these posteriors. The sample means for exact matching and Thompson sampling differ by one to two percent of cumulative regret in both regimes, which is replication noise. The VBOS policy’s mean regret is ten to eleven percent above Thompson sampling’s, and its action distribution is at total-variation distance 0.13 to 0.16 from the computed vector. F-LITE’s is four to five percent above. Per-replication regrets were not saved, so these differences carry no paired intervals.

The identification rates do not separate the policies. Their differences are within two binomial standard errors at 100 replications. The comparison is in the regret and distance columns.

7 Scope

The derivations of Section 3 are exact because the observations there are generated with factor structure. When scenario effects enter a simulation nonlinearly, a low-rank model of Λ is an approximation, and the posterior inherits its error. When a Gaussian-process kernel is dense with a short length scale, the intake fit to the grammar has to precede the computation, and its residual is the error.

The vector says how uncertain the choice of best is. It does not say how much one more measurement would reduce that uncertainty, which is what a budget allocator needs. That quantity is the knowledge gradient (Frazier et al., 2008), an expectation over what the posterior

means will become after the measurement, and [Tuisov \(2026\)](#) treats it as the myopic value of computation. The computation here does not supply it.

Three one-step criteria should be kept apart. Reducing the entropy of p is entropy search. Raising $\max_i p_i$ aims at correct identification under zero-one loss, where choosing a near-identical runner-up counts as an error. Both are objectives of an optimizer rather than of a bandit, a distinction [Hennig and Schuler \(2012\)](#) draw: an optimizer infers the minimum whatever the values at its evaluation points, and a bandit is scored on those values.

The knowledge gradient raises the expected value of the final choice and cares about the size of the differences. The vector supplies the first two directly. An allocator that spends where the entries of p are close is a heuristic for the second, not a criterion.

The regret analysis of [O’Donoghue and Lattimore \(2021\)](#) depends on the VBOS tilt. Its measured distance from p quantifies the approximation, but does not by itself make a case against the tilt.

Thompson sampling needs one posterior sample and no vector, and the bandit experiment can be run that way. The computed vector matters where a sample is not enough: stopping, allocation, constraints and derandomization. The stopping problem is of that kind.

The stopping experiment’s loadings have rank at most one, and in the aligned regime the centered loadings are zero, so the exact rule faces an independent contest there while the marginal rules do not know it.

8 Discussion

The computed vector replaces two kinds of approximation, the marginal shortcuts and counted posterior draws, and the gain over each is different. Against the marginal shortcuts it does not depend on rank. Their error can approach one half with every marginal held fixed, and in the stopping experiment they spent 1.5 to 3.4 times the replicates at the same threshold. Thompson sampling itself is not at fault, since it uses the correlation. The loss is confined to the side that computes probabilities, and that is the side the computed vector replaces.

How much better the computed numbers are than counted draws depends on the factor rank. On the posteriors of the two experiments, where the centred loadings have rank at most two, one evaluation of a hundred alternatives takes about ten milliseconds and is accurate to 10^{-6} per entry. Counting draws for ten milliseconds gives about twenty thousand draws, and an entry near 0.01 is then known to seven percent of itself. Matching 10^{-6} by counting would take 10^{11} draws. At factor ranks three to five, on single hundred-arm posteriors of a linear sampler, one evaluation takes 0.3 to 1 second and 200,000 draws take 0.3 second, with errors of the same order. The advantage over counting is a low-rank advantage, and centring the loadings is what decides the rank.

Bayesian optimization has the same object. Entropy search ([Hennig and Schuler, 2012](#)) takes the posterior probability that the minimum lies at each of n representer points as the quantity an optimizer should learn, and scores a belief by its relative entropy from a uniform base measure. On the representer points that probability is Equation (1) in the minimum convention, a Gaussian integral over a polyhedral cone, which [Hennig and Schuler \(2012\)](#) call intractable after [Plackett \(1954\)](#). They approximate it by expectation propagation at a cost of $O(n^4)$ per vector, which limits n to well below a thousand, and their implementation uses fifty representer points and about ten seconds per choice of evaluation point. Their alternative, sampling functions and recording each minimum, costs $O(Sn^3)$ for S samples and has no derivatives.

For a factor-structured belief the computation here gives the same vector in time linear in n , so thousands of representer points cost milliseconds. The price is the covariance class. Expectation propagation takes any dense covariance. A dense kernel on the representer points has to be fitted to the grammar of factors, blocks, nested groups and trees before the computation here applies. That fit is the dense-covariance intake of Cotton (2026), and its residual is the error.

Entropy search also uses the derivatives of the vector with respect to the mean and covariance for a first-order expansion of the expected information gain. The computation here supplies the exact Jacobian with respect to the means, and not the covariance derivatives.

The innovation formulas of Hennig and Schuler (2012) give the hypothetical posterior after a measurement for any Gaussian process, and the closure of Section 3 is the case in which that posterior stays in factor form. The entropy-search acquisition $\alpha(a) = H(p) - \mathbb{E}_{Y_a|\mathcal{D}}[H(p^{Y_a})]$ then needs the closure, the computation, and a scalar quadrature over the measurement outcome. The scope is finite candidates and factor beliefs.

The offline evaluation used for those samplers, the replay method of Li et al. (2011), needs a uniformly random log because the sampler’s own propensities were not available, and it discards every logged event whose random arm differs from the policy’s choice, about all but one in K for K arms. With the propensities computed, the production log itself becomes the evaluation set.

References

- S. Agrawal and N. Goyal. Thompson sampling for contextual bandits with linear payoffs. In *ICML*, 2013.
- O. Chapelle and L. Li. An empirical evaluation of Thompson sampling. In *Advances in Neural Information Processing Systems 24*, 2011.
- P. Cotton. Scalable inversion of contests with correlated performances, including softmax and multinomial probit. arXiv:2609.01133, 2026. Also SSRN 7307363.
- P. I. Frazier, W. B. Powell, and S. Dayanik. A knowledge-gradient policy for sequential information collection. *SIAM Journal on Control and Optimization*, 47(5):2410–2439, 2008.
- P. Hennig and C. J. Schuler. Entropy search for information-efficient global optimization. *Journal of Machine Learning Research*, 13:1809–1837, 2012.
- L. Li, W. Chu, J. Langford, and X. Wang. Unbiased offline evaluation of contextual-bandit-based news article recommendation algorithms. In *WSDM*, 2011.
- N. Menet, J. Huebotter, P. Kassraie, and A. Krause. Efficiently estimating Gaussian probability of maximality. In *AISTATS*, 2025. arXiv:2501.13535.
- N. Menet, A. Terzić, M. Hersche, A. Krause, and A. Rahimi. Thompson sampling via fine-tuning of LLMs. In *ICLR*, 2026. arXiv:2510.13328.
- B. O’Donoghue and T. Lattimore. Variational Bayesian optimistic sampling. In *Advances in Neural Information Processing Systems 34*, 2021. arXiv:2110.15688.
- R. L. Plackett. A reduction formula for normal multivariate integrals. *Biometrika*, 41(3–4):351–360, 1954.

- J. Tarbouriech, T. Lattimore, and B. O'Donoghue. Probabilistic inference in reinforcement learning done right. In *Advances in Neural Information Processing Systems 36*, 2023. arXiv:2311.13294.
- M. Titsias. Variational learning of inducing variables in sparse Gaussian processes. In *AISTATS*, 2009.
- A. Tuisov. Search as computation allocation. arXiv:2607.27871, 2026.